AI Lab Task # 03

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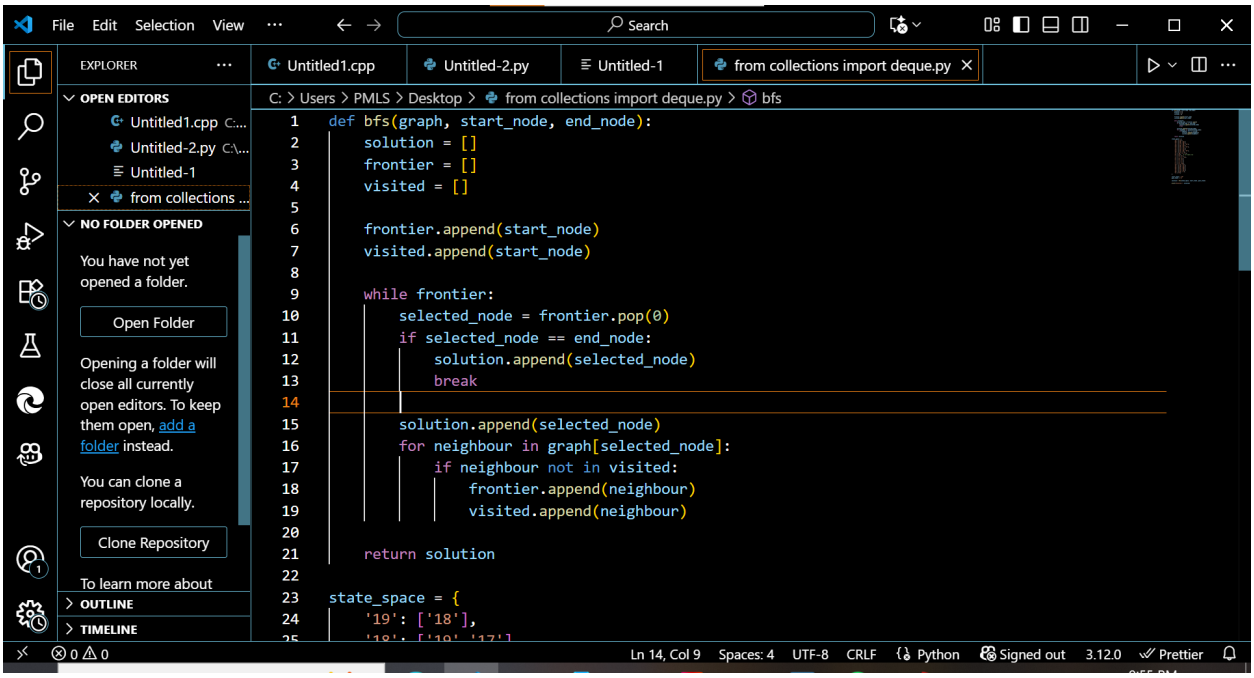
Section	B

Task# 01



```python
    return solution

state_space = {
    '19': ['18'],
    '18': ['19','17'],
    '17': ['18','16'],
    '16': ['17','20','15'],
    '15': ['16','13'],
    '13': ['15','12','14'],
    '12': ['13','11'],
    '14': ['13','11'],
    '11': ['12','14','10'],
    '10': ['11','8'],
    '8': ['10','7','9'],
    '9': ['8'],   # not empty now
    '7': ['8','6'],
    '6': ['7','5','G'],
    '5': ['6','4'],
    '4': ['5','3'],
    '3': ['4','2'],
    '2': ['3','1'],
    '1': ['2','25'],
    '20': ['16','21'],
    '21': ['20','22'],
```

OUTPUT

```python
def bfs(graph, start_node, end_node):
    solution = []
    frontier = []
    visited = []

    frontier.append(start_node)
    visited.append(start_node)
```

```
', '24', '10', '25', '8', '1', '7', '9', '2', '6', '3', '5', 'G']
PS C:\Users\PMLS>
```